EPS Rev B — integrated development schematic

Checkpoint: controller, both converters, solar and rail protection. Physical RBF, charger PowerPath and BMS remain absent.

Controller

File: controller.kicad_sch

CAN interfaces

File: can_interfaces.kicad_sch

Essential 3V3

File: essential_3v3.kicad_sch

Recovery 5V converter

File: recovery_5v.kicad_sch

Reference ranges: 100 controller, 200 BMS, 300 charger, 400 essential 3V3, 500 recovery 5V, 600 protection.

Solar source protection

File: source_protection.kicad_sch

Recovery rail protection

File: rail_protection.kicad_sch

SOLAR_PERMITTED -> source barrier -> VIN_CHG for FUTURE charger. VIN_CHG is NOT joined to SYS_PWR_PERMITTED.

SYS_PWR_PERMITTED remains an independent external permitted-source fixture. REG -> rail barrier -> protected+5V.

Incomplete ERC / low-VCC accuracy / hot SOA / shared startup /3V3 OV remain open. S03/S08/S11/S04 API limits; no PCB sync.

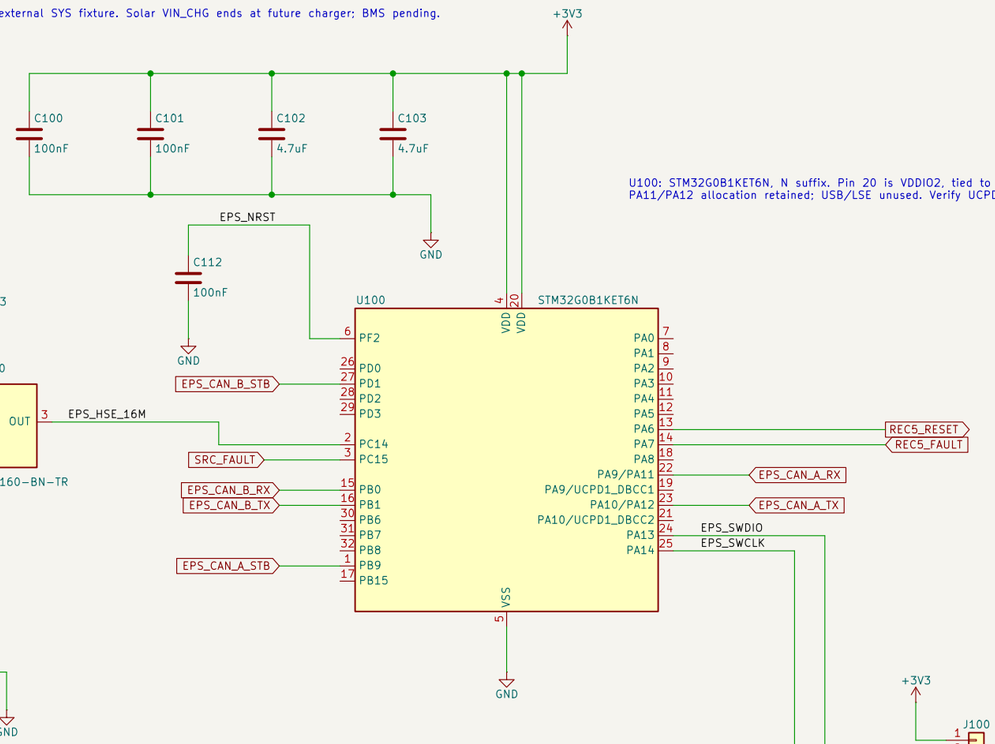
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CONTROLLER — integrated development checkpoint

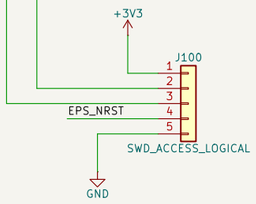
Essential+3V3 comes from buck; converters use external SYS fixture. Solar VIN_CHG ends at future charger; BMS pending.



16 MHz HSE bypass; no passive crystal.
Place C110/C111 locally at Y100 in layout; source-series option open.

U100: STM32G0B1KET6N, N suffix. Pin 20 is VDDIO2, tied to VDD.
PA11/PA12 allocation retained; USB/LSE unused. Verify UCPD reset configuration.

PROTECTION INTERFACES — LINKED GLOBALS
PA6 REC5_RESET ->47k R623 ->Q609; existing100k base pull-down.
PA7 REC5_FAULT; PC15 SRC_FAULT:active-low, single47k pullups on protector sheets.
Reset2ms; >=10s cooldown plus real rail qualification and settling.
Reset never bypasses hardware protection.



J100: logical SWD access; physical connector unselected.
1 VTref sense only, 2 SWDIO, 3 SWCLK, 4 NRST, 5 GND.
No injected probe power. Power-off/backfeed review open.

OPEN: ADC conditioning, local charger I2C, isolated pack SMBus,
UART, interlock status and payload request. Required pins remain unconnected.
No EPS drive on IHU-owned COMMS_EN / PAYLOAD_EN or stack alert.

PD3 reserved. Do not connect BMS BTP directly to 3V3 logic.
Unused BTP is tied to BMS VSS in the proposed BMS circuit; poll pack status.

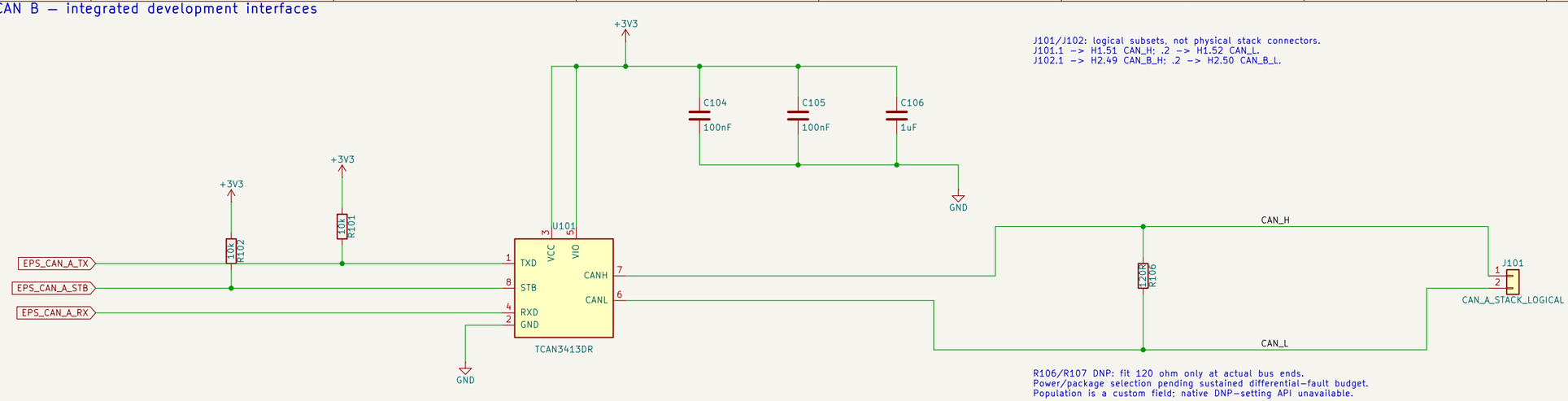
S11 OPEN: metadata is saved; required visible field styling unsupported.
S08: reused TCAN/ECS custom symbols remain under original construction review.

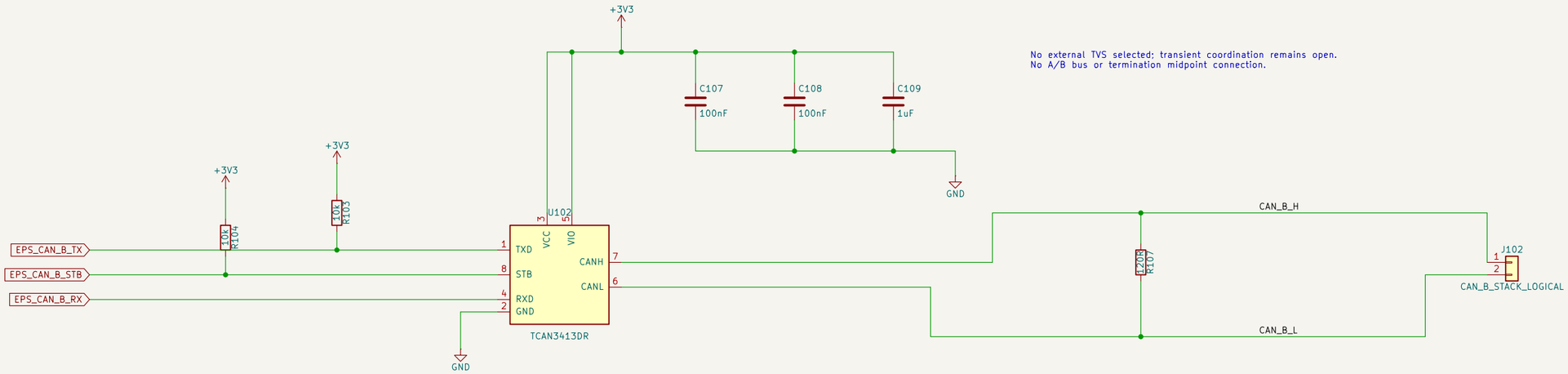
Sheet: /Controller/
File: controller.kicad_sch

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CAN A / CAN B — integrated development interfaces





FAULT LIMIT: shared +3V3, GND, MCU and clock remain common paths.
Independent VCC/VIO + pullup branch limiting remains a written candidate.
CAN B is bus-path recovery, not independent controller/power redundancy.

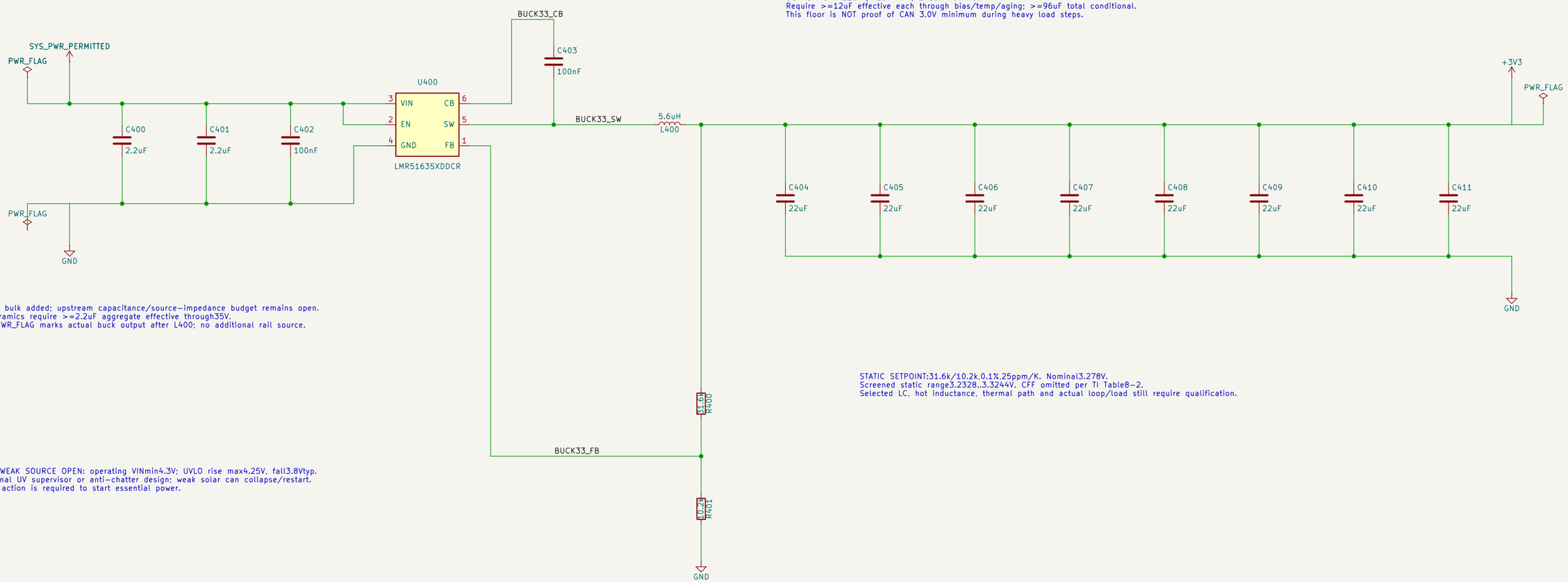
Existing TCAN custom symbols and manufacturer patterns reused.
Pin maps verified; assembly, ESD, timing and physical placement not qualified.
S08 construction review / S11 field-display limitations remain open.

Sheet: /CAN interfaces/		
File: can_interfaces.kicad_sch		
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ESSENTIAL 3V3 — LMR51635XDDCR INTEGRATED DEVELOPMENT

EXTERNAL FIXTURE: SYS_PWR_PERMITTED and GND represent a bounded permitted source.
Input/return PWR_FLAGs do not implement source isolation, RBF, charger or battery protection.

5..35 V development envelope; 4.5 V sensitivity. EN = VIN; starts without MCU.
2 A is steady sizing only. Actual MCU/IHU/CAN ramp/load-step budget and transient acceptance remain open.



No input bulk added; upstream capacitance/source-impedance budget remains open.
Input ceramics require >=2.2uF aggregate effective through35V.
Output PWR_FLAG marks actual buck output after L400; no additional rail source.

OUTPUT: 8 x 22uF / 25V X7R, 176uF nominal.
Require >=12uF effective each through bias/temp/aging; >=96uF total conditional.
This floor is NOT proof of CAN 3.0V minimum during heavy load steps.

STATIC SETPOINT:31.6k/10.2k,0.1%,25ppm/K. Nominal3.278V.
Screened static range3.2328..3.3244V. CFF omitted per TI Table8-2.
Selected LC, hot inductance, thermal path and actual loop/load still require qualification.

UVLO / WEAK SOURCE OPEN: operating VINmin4.3V; UVLO rise max4.25V, fall3.8Vtyp.
No external UV supervisor or anti-chatter design; weak solar can collapse/restart.
No MCU action is required to start essential power.

OPEN AUTONOMOUS 3V3 OV CONTAINMENT: failed-short high-side can expose MCU/CAN
and25V output bank to-35V. Upstream solar OV protection does not contain this fault.
This stage does not establish complete EPS essential-power acceptance.

POLICY OPEN: S03 native severity and S11 field-display APIs unavailable.
Existing custom U400 reused; S08 original review-marker coverage unresolved. No PCB synchronization.

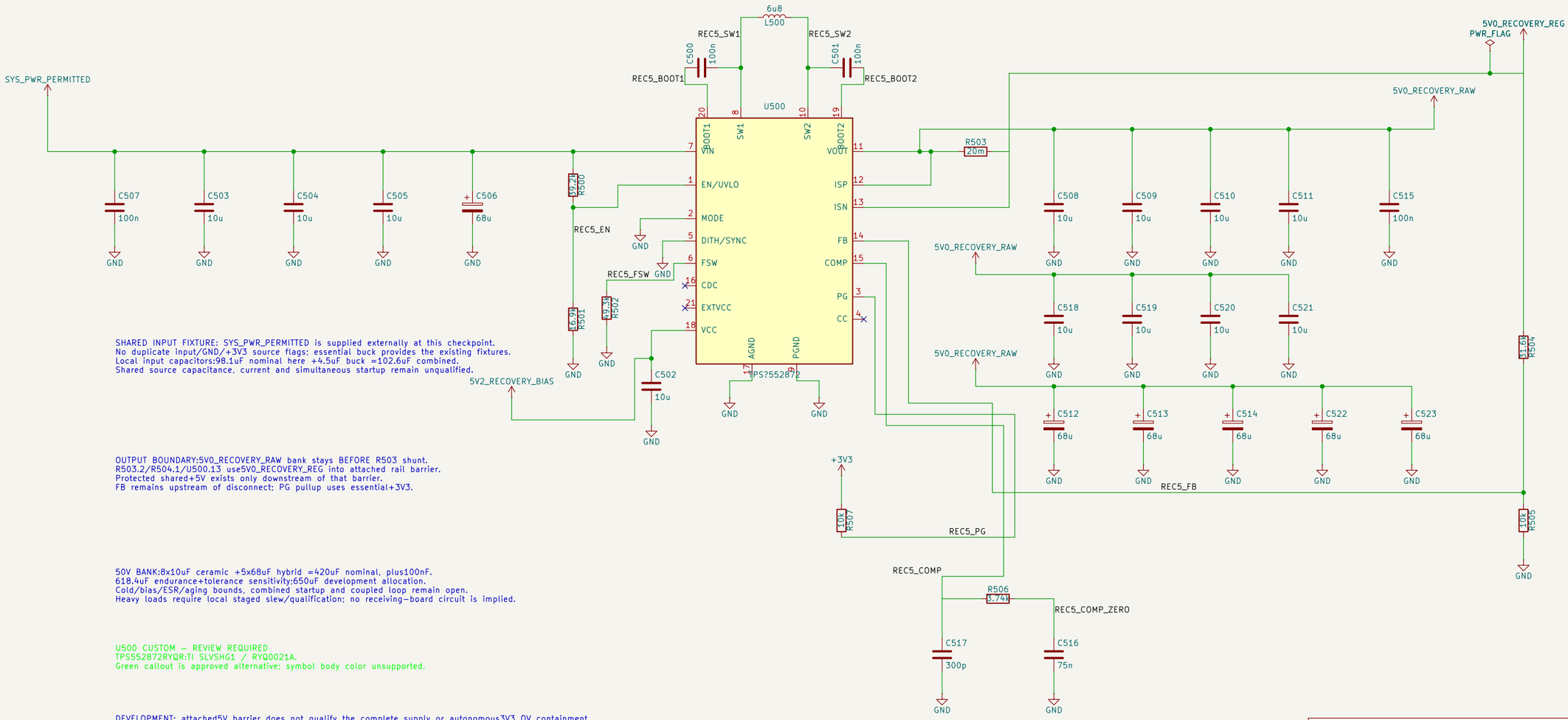
Sheet: /Essential 3V3/
File: essential_3v3.kicad_sch

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RECOVERY 5V CONVERTER — INTEGRATED DEVELOPMENT

TPS552872RYQR: autonomous EN/UVLO, internal bias via floating EXTVCC; MODE low (PFM).
4.5..35 V development envelope: 2 A steady target only. Source/thermal/dynamic qualification remains open.



SHARED INPUT FIXTURE: SYS_PWR_PERMITTED is supplied externally at this checkpoint.
No duplicate input/GND/+3V3 source flags; essential buck provides the existing fixtures.
Local input capacitors:98.1uF nominal here +4.5uF buck =102.6uF combined.
Shared source capacitance, current and simultaneous startup remain unqualified.

OUTPUT BOUNDARY:5V0_RECOVERY_RAW bank stays BEFORE R503 shunt.
R503.2/R504.1/U500.13 use5V0_RECOVERY_REG into attached rail barrier.
Protected shared+5V exists only downstream of that barrier.
FB remains upstream of disconnect; PG pullup uses essential+3V3.

50V BANK:8x10uF ceramic +5x68uF hybrid =420uF nominal, plus100nF.
618.4uF endurance+tolerance sensitivity:650uF development allocation.
Cold/bias/ESR/aging bounds, combined startup and coupled loop remain open.
Heavy loads require local staged slew/qualification; no receiving-board circuit is implied.

U500 CUSTOM — REVIEW REQUIRED
TPS552872RYQR:TI SLVSHG1 / RYQ0021A.
Green callout is approved alternative; symbol body color unsupported.

DEVELOPMENT: attached5V barrier does not qualify the complete supply or autonomous3V3 OV containment.
SYS remains an external permitted fixture; physical RBF/charger/BMS absent. No PCB synchronization.

Sheet: /Recovery 5V converter/		
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SOLAR SOURCE:33.5V analog clamp, timed cutoff, automatic retry

DEVELOPMENT ONLY. SOLAR_PERMITTED is an external interlocked-solar fixture; physical RBF is not drawn.

L600 limits input-short slew:require >=2.5uH through fault current.
C608=10nF timer. Bias/OV/snubber follow SRC_FILTERED.
Hot SOA, actual source energy and magnetic qualification remain open.

SOLAR_PERMITTED
PWR_FLAG
4.7u
L600
SRC_FILTERED
PWR_FLAG
Q600
IRLR2908
SRC_GH
SRC_DG
SRC_COMMON
IRLR2908
Q601
SRC_SENSE
R600
20m
PWR_FLAG
VIN_CHG
SMBJ60A
D604
GND
C600
68u
C601
10u
C602
10u
C603
10u
GND

SRC_FILTERED
R601
100
1N4148W-7-F
D600
PWR_FLAG
SRC_BIAS
C604
2.2u
C605
2.2u
C606
2.2u
C607
2.2u
GND

HGATE support / gate-source clamps / controlled startup

SRC_HGATE
SRC_COMMON
SRC_DG
SRC_SENSE
VIN_CHG
SRC_BIAS
U600
VCC
GATE
SOURCE
DGATE
SENSE
OUT
SHDN
FB
SRC_UV
SRC_OV
UV
OV
TMR
GND
GND
ENOUT
FLT
LTC4364HDE-2
SRC_FB
VIN_CHG
25.5k
R606
+3V3
47k
R608
SRC_FAULT
1k
R607
C608
10n
GND

1N4148W-7-F
D601
SRC_HGATE
R609
4.7k
SRC_GH
MMBT3906LT1G
Q602
100
R610
SRC_COMMON

SRC_GH
SMBJ8.5A
D602
SRC_COMMON
SRC_DG
SMBJ8.5A
D603
SRC_COMMON

SRC_FAULT:active-low global to MCU PC15; single47k R608 pullup to essential+3V3.
SHDN and ENOUT intentionally NC; no MCU startup dependency.

SRC_BIAS
50k
R602
SRC_UV
C609
10n
10k
R603
GND

SRC_FILTERED
27.8k
R604
SRC_OV
C610
1n
1k
R605
GND

SRC_GH
C611
1n
2.2k
R611
GND

VIN_CHG is protected solar output for FUTURE charger PowerPath.
It is NOT joined to converter SYS_PWR_PERMITTED in this schematic.
Source output bank98uF nominal; downstream/source energy remains unqualified.

PV fixture <=1A; raw reservoir <=1uF;
harness<=1uH:60V abnormal test only.
SMBJ60A is not a blanket80V clamp.
Snubber>=2uF effective through60V;
actual source energy/hot leakage open.

SRC_FILTERED
R612
1.0
SRC_SNUB
C612
2.2u
C613
2.2u
C614
2.2u
C615
2.2u
GND

S03 native severity /S11 field presentation APIs remain open. Existing Panasonic review candidate reused.
No hot-SOA, arbitrary-source, physical-fit or complete EPS power qualification.

Sheet: /Solar source protection/
File: source_protection.kicad_sch
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RECOVERY RAIL:5.270V analog clamp, timed latch-off

DEVELOPMENT: low-VCC FB accuracy, hot SOA and actual loads remain unqualified.

5V0_RECOVERY_REG
REC5_PIN
PWR_FLAG
4.7u
L601
Q603 IRLR2908
Q604 IRLR2908
Q605 IRLR2908
REC5_GH
REC5_DG
IRLR2908 Q606
IRLR2908 Q607
IRLR2908 Q608
REC5_COMMON
REC5_SENSE
R613 20m
+5V
PWR_FLAG
C616 100u
C617 100u
C618 22u
C619 22u
C620 22u
C621 22u
C622 22u
C623 22u
C624 22u
C625 22u
330 R625
GND

HGATE support / gate-source clamps / controlled startup

Three FETs per function reduce loss; equal linear-fault sharing is not assumed.

REC5_PIN
C627 2.2u
C628 2.2u
C629 2.2u
C630 2.2u
SMBJ36A D608
GND

REC5_HGATE
REC5_COMMON
REC5_DG
REC5_SENSE
+5V
REC5_BIAS
U601
VCC
GATE
SOURCE
DGATE
SENSE
OUT
SHDN
FB
UV
OV
TMR
GND
ENOUT
FLT
LTC4364HDE-1
3.57k R617
REC5_FB
REC5_UV
REC5_FAULT
1.11k R618
+3V3
47k R619
C632 100n

1N4148W-7-F D605
REC5_HGATE
R620 4.7k
REC5_GH
MMBT3906LT1G Q610
100 R621
REC5_COMMON

SMBJ8.5A D606
SMBJ8.5A D607
REC5_GH
REC5_DG
REC5_COMMON

REC5_PIN
R614 100
PWR_FLAG
REC5_BIAS
C631 100n
REC5_BIAS
20k R615
REC5_UV
10k R616
Q609 MMBT3904LT1G
REC5_RESET
R623 47k
100k R624

REC5_FAULT:active-low global to MCU PA7; single47k R619 pullup to essential+3V3.
SHDN/ENOUT intentionally NC. Reset PA6 acts through47k R623 and Q609.

Protected bank:2x100uF hybrid+8x22uF ceramic nominal.
Require>=112uF bulk /<=0.4ohm cold and>=48uF low-ESR ceramic;external<=100uF.
Actual cap/magnetic/pulse/source bounds and consumer voltage-time limits remain open.

5V0_RECOVERY_REG is actual converter output; converter FB and RAW bank stay UPSTREAM.
R625=330ohm bleed; C632=100nF timer; C626=10nF gate slew; R617/R618=3.57k/1.11k.
Require5ms heavy-load admission plus20ms settling and local rail qualification.
2ms UV reset never bypasses cooldown:>=10s development wait; restore<=0.5A startup state.

INTEGRATION IS NOT QUALIFICATION. Low-VCC clamp accuracy and hot linear SOA remain conditional.
Autonomous3V3 OV containment is absent. S03/S11 API limitations and reused-part review gates remain open.

C626 10n
2.2k R622

Sheet: /Recovery rail protection/		
File: rail_protection.kicad_sch		
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